

Notes: Chernenko, Hanson & Sunderam (JFE, 2016)

Who Neglects Risk? Investor Experience and the Credit Boom

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Mutual fund holdings and manager characteristics	4
3.2	Nontraditional securitization share	4
3.3	Manager experience and inexperience	4
3.4	Riskiness of NTS holdings	5
3.5	Personal experience variables	5
4	Models and empirical specifications	5
4.1	Baseline cross-sectional holdings regression	5
4.2	Riskiness regression	5
4.3	Evolution and local house price experience	6
4.4	Past investment outcomes and 1998 experience	6
4.5	Crisis trading and alternative explanations	6
5	Identification and empirical design	6
6	Section-by-section study guide	7
7	Figures and tables	8
7.1	Figure 1: Issuance and credit spreads on traditional and nontraditional securitizations	8
7.2	Figure 2 and Table 1: Manager experience and summary statistics	9
7.3	Table 2: Impact of experience on 2007 NTS holdings	9
7.4	Table 3: Robustness - impact of experience on 2007 NTS holdings	10
7.5	Table 4 and Figure 3: Riskiness and evolution of NTS holdings	10
7.6	Table 5 and Figure 4: Evolution over time and local house price appreciation	11
7.7	Table 6: Prior manager return experiences and 2007 NTS holdings	12
7.8	Table 7 and Figure 5: 1998 credit-market experience	13
7.9	Table 8: Institutional memory and manager training	13
7.10	Table 9: Trading by inexperienced managers during the crisis	14
7.11	Figure 6 and Table 10: Performance-flow incentives, returns, and alternative explanations	14
8	Detailed result map	15
9	Short summary	15

10 Conclusion	16
10.1 Core conclusion	16
10.2 Incremental contribution revisited	16
10.3 Policy implication	16
10.4 Limitations	16
10.5 Takeaway	16

1 Big picture

- **Question.** Which investors bought the risky credit products at the center of the 2003–2007 mortgage credit boom, and why did they appear to neglect downside risk?
- **Setting.** The paper uses micro-data on mutual funds’ fixed-income holdings from 2003 to 2010. Mutual funds are useful because manager tenure, portfolio holdings, flows, and past returns can be measured at the fund-manager level.
- **Main assets.** The paper studies nontraditional securitizations (NTS): private-label nonprime RMBS and CDOs. These were relatively new, had short performance histories, and were central to the financial crisis.
- **Main behavioral mechanism.** Beliefs are shaped by firsthand experience. Inexperienced managers extrapolated from the quiet boom years, neglected rare left-tail credit risk, and therefore found NTS attractive. Seasoned managers had observed more credit-market stress and were less bullish.
- **Main cross-sectional result.** In 2007, inexperienced managers held roughly twice as much NTS as seasoned managers. The baseline estimate is about four percentage points of additional NTS share for inexperienced managers, economically large relative to the sample mean.
- **Risk-taking margin.** Inexperienced managers not only held more NTS; they also held riskier NTS, including higher-spread and lower-rated tranches.
- **Mechanism evidence.** The effects are stronger when managers’ own local house-price experiences were favorable, when managers avoided severe adverse investment outcomes, and when they did not experience the 1998 credit-market disruption.
- **Substitutes for personal experience.** Institutional memory inside fund families and formal training, especially CFA credentials, partly substitute for personal experience and reduce NTS exposure.
- **Crisis behavior.** Once the crisis began, inexperienced managers sold more NTS and held NTS that were written down more severely, consistent with a larger revision of previously optimistic beliefs.
- **Economic takeaway.** Credit booms can be fueled by path-dependent beliefs. When market participants have not personally experienced bad outcomes, they may neglect rare disaster risk and demand too much risky credit.

2 Incremental contribution of the paper

- **From credit-boom narratives to micro evidence.** Many crisis narratives argue that lenders became overly optimistic during the mortgage boom. This paper turns that narrative into testable micro-level evidence using individual mutual fund managers’ experience and holdings.
- **Experience effects in professional credit markets.** Prior work, such as Greenwood and Nagel’s work on technology stocks, studies inexperienced investors in equity bubbles. Chernenko, Hanson, and Sunderam show that the same experience channel matters in professional fixed-income markets and in complex credit products.
- **Focus on the absence of negative experience.** Earlier experience-effects papers often emphasize how bad experiences make investors more conservative. This paper’s novelty is the mirror image: investors who *lack* bad experiences can become overoptimistic and neglect downside risk.
- **Asset-class-specific belief distortion.** The paper does not simply document generic risk seeking by young managers. The effect is specific to nontraditional securitizations, and not to traditional securitizations or GSE MBS, supporting a belief distortion about new structured-credit products.
- **Mechanism through personal and local experiences.** The paper connects manager behavior to (i) tenure, (ii) local house price appreciation, (iii) personal portfolio outcomes, (iv) experience during the 1998 credit disruption, and (v) family-level institutional memory.

- **Institutional memory and training.** The paper adds an organizational layer to behavioral finance. Fund families and CFA training can transmit cautionary knowledge and partially substitute for firsthand experience.
- **Dynamic boom-to-bust evidence.** The paper follows holdings through the boom and crisis. It shows that inexperienced managers increased NTS exposure during the boom, then sold more aggressively during the crisis.
- **Policy contribution.** The conclusion reframes losses as belief-updating devices. Moderate losses may help preserve financial memory and reduce the buildup of overoptimistic beliefs before a crisis.

3 Data and measurement

3.1 Mutual fund holdings and manager characteristics

- The data cover domestic taxable bond mutual funds and hybrid stock/bond mutual funds from 2003 through 2010.
- The paper excludes money market funds, index bond funds, and Treasury-only government bond funds.
- Holdings of securitizations are measured using eMAXX and related fixed-income holdings data.
- Fund-manager biographies from Morningstar provide manager start dates and fund tenures. For team-managed funds, manager characteristics are averaged across managers.
- The main experience measure is years since the manager first appears managing a fund in Morningstar, fixed as of 2004Q4 for the baseline.

Interpretation: Experience is measured before the main boom-time 2007 holdings outcome, reducing concern that funds hired young managers because they already planned to buy securitizations.

3.2 Nontraditional securitization share

The main portfolio variable is

$$NTS\ share = \frac{\text{par holdings of nonprime RMBS and CDOs}}{\text{fixed-income par holdings}}.$$

Traditional securitizations include CMBS, prime RMBS, consumer ABS, and other ABS. GSE MBS are treated separately because government guarantees make their credit risk different.

Interpretation: The NTS share captures exposure to the private-label mortgage and CDO products that were central to the credit boom and bust.

3.3 Manager experience and inexperience

- The baseline inexperience indicator equals one if the fund’s management team has below-median experience as of 2004Q4.
- A continuous inexperience variable is defined as negative years of experience.
- Later robustness checks use 2007 experience and alternative cutoff years.

Interpretation: The paper separates chronological age from relevant investment experience. In untabulated results, manager age has much weaker explanatory power than fund-management experience.

3.4 Riskiness of NTS holdings

- Security-level new-issue spreads are used to measure the riskiness of NTS held by each fund.
- Initial credit ratings are also used to distinguish AAA and non-AAA holdings.
- Table 4 tests whether inexperienced managers chose higher-spread NTS.

Interpretation: If inexperienced managers were merely following broad mandates, they might hold more of all securitizations. If they were overoptimistic about downside risk, they should tilt toward riskier tranches.

3.5 Personal experience variables

- Local house price appreciation is measured at the manager’s MSA from 2003 to 2006.
- Personal return histories from 1995 to 2005 measure the manager’s average return and worst return.
- Distant minimum return measures how far in the past the manager’s worst experience occurred.
- 1998 experience measures whether the manager was managing during the Russia/LTCM credit-market turmoil, and the manager’s personal 1998 outcome.

Interpretation: These variables allow the paper to distinguish mere years on the job from the content and timing of firsthand experiences.

4 Models and empirical specifications

4.1 Baseline cross-sectional holdings regression

The main 2007 holdings regression is

$$y_i = \alpha_{objective(i)} + \beta Inexperienced_i + \gamma' x_i + \epsilon_i,$$

where y_i is either $NTS\ share_i$ or an indicator for whether the fund holds any NTS. Controls include CFA status, team management, fund and family size, fund and family age, and family taxable-bond share.

Interpretation: The coefficient β measures whether inexperienced managers held more NTS than seasoned managers within the same fund objective and conditional on observable fund/family characteristics.

4.2 Riskiness regression

$$Spread_i = \alpha_{objective(i)} + \beta Inexperienced_i + \gamma' x_i + \epsilon_i.$$

The dependent variable is the par-weighted average new-issue spread of the NTS held by the fund.

Interpretation: A positive β indicates that inexperienced managers not only bought more NTS, but bought riskier NTS.

4.3 Evolution and local house price experience

The time-series cross-sections use year-specific regressions:

$$NTS\ share_{i,t} = \alpha_{objective(i)} + \beta_t Inexperienced_i + \gamma'_t x_{i,t} + \epsilon_{i,t}.$$

The local-experience test is

$$NTS\ share_i = \alpha + \beta_1 Inexperienced_i + \beta_2 High\ local\ HPA_i + \beta_3 (Inexperienced_i \times High\ local\ HPA_i) + \gamma' x_i + \epsilon_i.$$

Interpretation: The paper tests not just whether inexperienced managers bought more NTS, but whether the gap widened during the boom and was stronger for managers exposed to high local housing gains.

4.4 Past investment outcomes and 1998 experience

$$NTS\ share_i = \alpha + \beta_1 Inexperienced_i + \beta_2 HighAvgReturn_i + \beta_3 HighMinReturn_i + \beta_4 DistantMinReturn_i + \gamma' x_i + \epsilon_i.$$

The 1998 analysis uses indicators for whether the manager had started by cutoff years and whether the manager's own 1998 outcome was high.

Interpretation: This specification tests whether all experience is equal. The paper finds that good, recent, and tranquil experiences lead managers to buy more NTS.

4.5 Crisis trading and alternative explanations

The crisis-period trading regression is

$$y_{i,t} = \alpha_{objective(i),t} + \beta_1 Inexperienced_i + \beta_2 Flows_{i,t} + \beta_3 NTS\ share_{i,t-1} + \gamma' x_{i,t} + \epsilon_{i,t}.$$

Table 10 also studies whether inexperienced managers faced stronger performance-flow incentives or had different returns.

Interpretation: These tests help rule out alternative explanations based solely on incentives, forced selling, risk preferences, or skill.

5 Identification and empirical design

- **Core identification problem.** Inexperienced managers may differ from seasoned managers in unobserved skills, mandates, incentives, or clientele. The paper therefore tests a full pattern of predictions, not just one coefficient.
- **Pre-determined experience.** Baseline inexperience is measured in 2004Q4, before peak NTS holdings, reducing reverse causality.
- **Fund objective fixed effects.** The main regressions compare managers within the same fund objectives, reducing mandate-driven confounding.
- **Controls.** CFA status, team management, fund size, family size, fund age, family age, and taxable-bond family share address manager and family characteristics.
- **Mechanism tests.** Local HPA, personal returns, 1998 experience, family outcomes, and CFA training all directly test the belief-formation mechanism.
- **Robustness tests.** Results survive TNA weighting, largest-fund samples, alternative denominators, rating/tranche decompositions, and 2007 experience measurement.
- **Alternative explanations.** The paper examines incentives, risk preferences, skill, and forced selling; none parsimoniously explain the full pattern.

6 Section-by-section study guide

- **Introduction.** Explains how firsthand experience may shape credit-boom optimism and summarizes the NTS facts.
- **Background and hypotheses.** Defines traditional and nontraditional securitizations; develops predictions from representativeness and availability heuristics.
- **Data.** Describes mutual fund holdings, manager experience, fund characteristics, NTS measures, and local/personal experience variables.
- **Baseline results.** Shows inexperienced managers held far more NTS in 2007 and riskier NTS.
- **Evolution and local HPA.** Shows divergence over the boom and stronger holdings among inexperienced managers in high-HPA locations.
- **Personal experiences.** Shows recent/worst past outcomes and 1998 experiences shape later NTS holdings.
- **Substitutes and crisis behavior.** Studies institutional memory, CFA training, crisis sales, returns, and alternative explanations.
- **Conclusion.** Argues that financial memory can fade quickly and that moderate losses may preserve sensible beliefs about risk.

7 Figures and tables

7.1 Figure 1: Issuance and credit spreads on traditional and nontraditional securitizations

Read:

- Panel A shows that nontraditional securitization issuance surged during the 2003–2007 boom.
- Panel B shows credit spreads on AAA nontraditional securitizations converging toward traditional securitization spreads before widening sharply in 2007.
- The figure sets up the paper’s main belief question: why did investors demand so much of a new and risky product whose spreads became increasingly compressed?

Detailed interpretation: The figure is the macro background for the entire paper. NTS issuance grew rapidly, and market prices suggested that investors were not demanding much compensation for additional risk. This is consistent with boom-time optimism and motivates looking at which investors bought these securities.

Economic message: The boom was not simply an increase in credit supply; it was also a shift in investor demand for new structured-credit products.

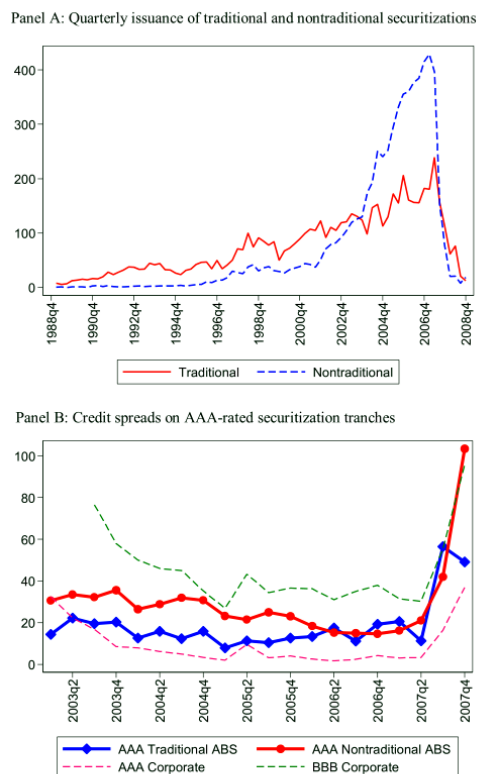


Fig. 1. Issuance and credit spreads on traditional and nontraditional securitizations. This figure shows quarterly issuance volume and credit spreads on traditional and nontraditional securitizations based on data

Figure 1: Issuance and credit spreads on traditional and nontraditional securitizations.

7.2 Figure 2 and Table 1: Manager experience and summary statistics

Read:

- Figure 2 shows that NTS holdings are highest for managers with short experience, especially those with 4–8 years of experience.
- Table 1 reports 5,983 fund-year observations, average manager experience of 8.52 years, and mean NTS share of 4.61 percent.
- The median NTS share is only 0.96 percent, showing that NTS exposure is highly right-skewed.
- The average fund has large GSE MBS exposure, but the paper focuses on nontraditional securitizations that carry more credit risk.

Detailed interpretation: Figure 2 gives the visual version of the main result: NTS holdings decline sharply with manager experience. Table 1 shows that NTS is not the whole fixed-income portfolio, but it is a meaningful and very skewed exposure.

Economic message: The paper studies a concentrated exposure to products at the center of the credit boom, and the exposure is disproportionately held by inexperienced managers.

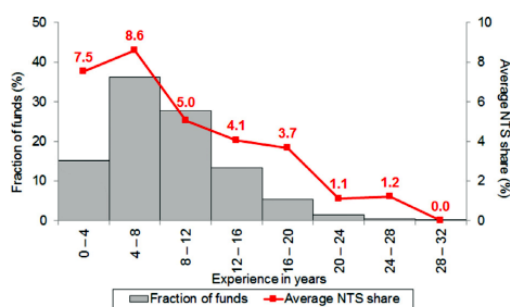


Fig. 2. Bond manager experience in years and 2007 average NTS holdings. The histogram shows the 2007 distribution of bond manager experience based on 4-year bins. Experience is measured as of 2004Q4. Experience is the number of years an individual has been managing mutual funds. For team-managed funds, we take the average of individual managers' experience.

(a) Figure 2: Manager experience and 2007 NTS holdings.

Table 1

Summary statistics.

This table reports summary statistics for our 2003 to 2010 annual panel of bond mutual funds. Our sample of mutual funds consists of domestic taxable bond funds and hybrid stock/bond funds, excluding money market funds, index bond funds, and Treasury-only government bonds funds. Specifically, we use funds with the following Lipper investment objective codes in CRSP: "A" (Corporate debt funds A-rated), "ARM" (Adjustable rate mortgage funds), "B" (Balanced funds), "BBB" (Corporate debt funds BBB-rated), "CA" (Capital appreciation funds), "FLX" (Flexible income funds), "FX" (Flexible portfolio funds), "GB" (General bond funds), "GNM" (GNMA funds), "GUS" (General U.S. government funds), "HY" (High yield funds), "I" (Income funds), "ID" (Intermediate investment grade debt funds), "ILG" (Intermediate U.S. government funds), "MSI" (Multi-sector income funds), "SID" (Short investment grade debt funds), "SIF" (Short-intermediate investment grade debt funds), "SIU" (Short-intermediate U.S. government funds), "SUS" (Short U.S. government funds), "USM" (U.S. mortgage funds), and "USO" (Ultra-short obligation funds). Our manager-level variables are from Morningstar and are measured as of 2004Q4. For team-managed funds, these variables reflect the average of each of the fund's managers. Experience is the number of years since we first observe each manager in Morningstar. CFA is the fraction of managers who are Chartered Financial Analysts. Fund TNA, Fund age, Family TNA, Family age, and Family taxable bond share are from the CRSP mutual fund database. Fund par fixed income assets is from eMaxx. NTS share, TS share, and GSE MBS share are portfolio shares as a fraction of par fixed-income holdings and are derived from eMAXX.

Variable	N	Mean	Median	St dev	Min	Max
Manager-level variables						
Experience (years)	5,983	8.52	7.87	4.51	0.00	29.06
CFA (fraction)	5,983	0.48	0.50	0.40	0.00	1.00
Number of managers	5,983	2.77	2.00	2.47	1.00	21.00
Team managed (indicator)	5,983	0.67	1.00	0.47	0.00	1.00
Fund-level variables						
Fund TNA (\$ billion)	5,983	1.56	0.30	7.06	0.00	236.62
Fund par fixed income assets (\$ billion)	5,983	1.07	0.23	4.14	0.00	175.96
Fund age (years)	5,983	15.90	13.63	11.86	0.15	83.80
Nontraditional securitization share(%)	5,983	4.61	0.96	9.03	0.00	100.00
Traditional securitization share(%)	5,983	10.04	5.15	13.34	0.00	93.89
GSE MBS share(%)	5,983	24.66	17.27	26.14	0.00	100.00
Family-level variables						
Family TNA (\$ billion)	5,983	123.52	29.20	262.56	0.00	1,683.2
Family age (years)	5,983	42.40	34.30	25.14	0.68	85.52
Family taxable bond share(%)	5,983	39.43	32.45	25.83	0.00	100.00

(b) Table 1: Summary statistics.

7.3 Table 2: Impact of experience on 2007 NTS holdings

Read:

- In Panel A, the inexperience dummy predicts roughly 3.4–3.9 percentage points more NTS share, depending on controls and fixed effects.
- Inexperienced managers are also 9–11 percentage points more likely to hold any NTS.
- In Panel B, continuous inexperience also predicts higher NTS holdings.
- CFA managers hold less NTS; team-managed funds hold more NTS.

Detailed interpretation: This is the baseline empirical table. The estimated effect is economically large relative to the 4.61 percent sample mean of NTS share. The result is robust to controls and investment objective fixed effects.

Economic message: The central empirical fact is that inexperience is strongly associated with buying boom-era nontraditional securitizations.

7.4 Table 3: Robustness - impact of experience on 2007 NTS holdings

Read:

- The baseline result survives TNA weighting and restriction to the 250 largest funds.
- It survives using NTS/TNA rather than NTS/bond holdings.
- The effect appears in non-AAA and nonprime RMBS/CDO decompositions.
- It does not appear in the same way for GSE MBS, which supports the asset-class-specific belief story.

Detailed interpretation: Table 3 rules out many mechanical interpretations of Table 2. The finding is not driven only by tiny funds, by denominator choice, or by general securitization exposure. It is strongest for the risky, crisis-relevant NTS categories.

Economic message: The overoptimism channel is specific to new risky credit products rather than a generic taste for all securitizations.

Table 2

Impact of experience on 2007 NTS holdings.

This table reports the results of cross-sectional regressions of bond mutual funds' 2007 holdings of nontraditional securitizations on their portfolio managers' experience and various fund characteristics:

$$y_i = \alpha_{\text{experience}} + \beta_1 \text{Inexperienced}_i + \gamma'X_i + \varepsilon_i$$

Panel A uses a dummy indicator equal to one for managers with below median experience as of 2004Q4, while Panel B shows the same result for a continuous measure of inexperience: $-1 \times$ years of experience. The dependent variable in columns 1 to 4 is the nontraditional share ($y_i = \text{NTS share}_i$), while the dependent variable in columns 5 to 8 is an indicator equal to one if the fund holds NTS. $\text{Has NTS}_i = 1(\text{NTS share}_i > 0)$. Fund objective fixed effects are included in columns 3, 4, 7, and 8. Robust t -statistics are shown in square brackets below the coefficient estimates.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Dummy indicator for inexperienced managers								
	$y_i = \text{NTS share}_i(\%)$				$y_i = \text{Has NTS}_i$			
Inexperienced dummy	3.932 [4.77]	3.560 [4.56]	3.467 [4.65]	3.376 [4.66]	0.097 [2.86]	0.090 [2.75]	0.101 [3.19]	0.107 [3.45]
CFA	-2.812 [-3.10]	-2.812 [-3.10]	-2.215 [-2.48]	-2.215 [-2.48]	-0.058 [-1.38]	-0.058 [-1.38]	-0.040 [-0.99]	-0.040 [-0.99]
Team managed	2.857 [3.82]	2.857 [3.82]	2.325 [3.41]	2.325 [3.41]	0.165 [4.53]	0.165 [4.53]	0.125 [3.63]	0.125 [3.63]
log(Fund TNA)	0.391 [1.23]	0.391 [1.23]	0.473 [1.56]	0.473 [1.56]	0.038 [2.94]	0.038 [2.94]	0.034 [2.89]	0.034 [2.89]
log(Family TNA)	0.003 [0.01]	0.003 [0.01]	-0.186 [-0.68]	-0.186 [-0.68]	0.029 [2.54]	0.029 [2.54]	0.029 [2.74]	0.029 [2.74]
log(Fund age)	-1.636 [-3.05]	-1.636 [-3.05]	-0.662 [-0.99]	-0.662 [-0.99]	-0.042 [-1.51]	-0.042 [-1.51]	-0.009 [-0.32]	-0.009 [-0.32]
log(Family age)	-0.773 [-0.91]	-0.773 [-0.91]	-0.574 [-0.73]	-0.574 [-0.73]	-0.021 [-0.66]	-0.021 [-0.66]	0.018 [0.59]	0.018 [0.59]
Family taxable bond share(%)	0.075 [4.15]	0.075 [4.15]	0.055 [3.18]	0.055 [3.18]	0.002 [2.75]	0.002 [2.75]	0.002 [2.72]	0.002 [2.72]
Fund objective fixed effects	No	No	Yes	Yes	No	No	Yes	Yes
Observations	757	757	757	757	757	757	757	757
R-squared	0.03	0.11	0.19	0.23	0.01	0.11	0.19	0.27
Panel B: Continuous measure of inexperience								
	$y_i = \text{NTS share}_i(\%)$				$y_i = \text{Has NTS}_i$			
Inexperience = $-1 \times$ years	0.355 [4.66]	0.277 [3.69]	0.283 [4.07]	0.255 [3.66]	0.014 [3.75]	0.013 [3.54]	0.015 [4.35]	0.016 [4.54]
CFA	-2.873 [-3.12]	-2.873 [-3.12]	-2.307 [-2.52]	-2.307 [-2.52]	-0.063 [-1.50]	-0.063 [-1.50]	-0.047 [-1.17]	-0.047 [-1.17]
Team managed	2.832 [3.62]	2.832 [3.62]	2.337 [3.28]	2.337 [3.28]	0.155 [4.20]	0.155 [4.20]	0.113 [3.23]	0.113 [3.23]
log(Fund TNA)	0.366 [1.13]	0.366 [1.13]	0.425 [1.39]	0.425 [1.39]	0.040 [3.09]	0.040 [3.09]	0.036 [3.03]	0.036 [3.03]
log(Family TNA)	0.042 [0.14]	0.042 [0.14]	-0.141 [-0.51]	-0.141 [-0.51]	0.029 [2.55]	0.029 [2.55]	0.029 [2.78]	0.029 [2.78]
log(Fund age)	-1.590 [-2.02]	-1.590 [-2.02]	-0.588 [-0.89]	-0.588 [-0.89]	-0.033 [-1.18]	-0.033 [-1.18]	-0.003 [-0.09]	-0.003 [-0.09]
log(Family age)	-0.854 [-1.00]	-0.854 [-1.00]	-0.671 [-0.85]	-0.671 [-0.85]	-0.024 [-0.76]	-0.024 [-0.76]	0.014 [0.49]	0.014 [0.49]
Family taxable bond share(%)	0.073 [3.97]	0.073 [3.97]	0.053 [3.01]	0.053 [3.01]	0.002 [2.66]	0.002 [2.66]	0.002 [2.65]	0.002 [2.65]
Fund objective fixed effects	No	No	Yes	Yes	No	No	Yes	Yes
Observations	757	757	757	757	757	757	757	757
R-squared	0.02	0.10	0.18	0.22	0.02	0.12	0.19	0.28

and economically significant: a fund whose family was all funds.¹⁷ Nonetheless, the fixed effects have little effect on

Table 2: Impact of experience on 2007 NTS holdings.

7.5 Table 4 and Figure 3: Riskiness and evolution of NTS holdings

Read:

- Table 4 shows that inexperienced managers held NTS with higher new-issue spreads, especially before controls.
- Figure 3 shows that both groups began around 3 percent NTS share in 2003, but inexperienced managers increased exposure much more during the boom.

Table 3

Robustness: Impact of experience on 2007 NTS holdings.

This table reports a battery of robustness exercises for our main result. Specifically, we report cross-sectional regressions of bond mutual funds' 2007 holdings of nontraditional securitizations on their portfolio managers' experience and various fund characteristics:

$$y_i = \alpha_{\text{experience}} + \beta_1 \text{Inexperienced}_i + \gamma'X_i + \varepsilon_i$$

For each robustness exercise, we show results measuring inexperience using an indicator for below-median experience and using our continuous measure of experience in years. We also show results for both our continuous holdings measure ($y_i = \text{NTS share}_i$) and an indicator for funds holding any NTS ($y_i = \text{Has NTS}_i$). Row 1 repeats our baseline results from Table 2. We first modify our sample weighting and subsample. Row 2 shows results weighting each fund by its total net assets (TNA). Row 3 shows results restricting attention to the 250 largest funds in our sample by TNA. We next modify our dependent variable. Row 4 shows results using $y_i = \text{NTS}/\text{TNA}$, in place of $y_i = \text{NTS}/\text{RWG}$. Rows 5 and 6 decompose NTS holdings into AAA holdings and non-AAA holdings based on Moody's credit ratings. (We classify tranches that receive a rating of "BB" or AAA because there are direct, though upper-order, tranches that were not rated. However, this makes little difference.) Thus, when the dependent variable is measured continuously, the sum of the coefficients in rows 5 and 6 equals the coefficient in row 1. Similarly, rows 7 and 8 decompose NTS holdings into Mortgage RMBS and CDO holdings. In rows 9 and 10, the dependent variable is the fraction of nontraditional securitizations (TV share) and GSE-backed MBS (GSE MBS share) in each fund's portfolio, respectively. Finally, we modify the inexperience variable. Specifically, row 11 shows the results when experience is defined as of 2007 instead of 2004. All specifications include the full suite of controls from Table 2 as well as fund objective fixed effects. Robust t -statistics are shown in square brackets.

	N	Inexperience dummy				Inexperience = -1 years			
		$y_i = \text{NTS share}_i(\%)$		$y_i = \text{Has NTS}_i$		$y_i = \text{NTS share}_i(\%)$		$y_i = \text{Has NTS}_i$	
		β	t	β	t	β	t	β	t
(1) Baseline	757	3.376	[4.66]	0.097	[2.86]	0.277	[3.69]	0.283	[4.07]
(2) TNA-weighted	757	3.498	[3.68]	0.39	[0.72]	0.37	[3.17]	0.37	[3.51]
(3) 250 largest funds	250	2.566	[2.95]	0.39	[0.79]	0.28	[3.01]	0.40	[3.20]
Moody dependent variable									
(4) NTS/TNA	757	4.75	[4.12]	0.17	[0.87]	0.27	[0.82]	0.16	[0.66]
(5) NTS _{AAA} share	757	2.46	[4.04]	0.21	[0.89]	0.34	[3.15]	0.21	[0.81]
(6) NTS _{non-AAA} share	757	0.81	[3.41]	0.08	[0.80]	0.21	[0.80]	0.08	[0.61]
(7) NTS _{AAA} share	757	2.86	[4.11]	0.24	[0.71]	0.34	[3.15]	0.21	[0.81]
(8) NTS _{non-AAA} share	757	0.52	[2.34]	0.05	[0.39]	0.13	[0.75]	0.06	[0.61]
(9) TV share	757	1.22	[1.31]	0.32	[0.79]	0.31	[0.97]	0.32	[0.99]
(10) GSE MBS share	757	-2.72	[-2.89]	0.57	[0.86]	0.37	[0.65]	0.57	[0.86]
(11) Moody independent variable	757	1.043	[1.41]	0.21	[0.96]	0.27	[0.34]	0.21	[0.91]

Table 3: Robustness tests.

- By 2007Q2, inexperienced managers held nearly twice the NTS share of seasoned managers.

Detailed interpretation: The table and figure together show that inexperience is linked to both the level and the timing of NTS exposure. Inexperienced managers did not always hold more NTS. The gap developed during the tranquil boom.

Economic message: Boom-time learning was asymmetric: the absence of bad outcomes caused inexperienced managers to become increasingly optimistic.

Table 4

Impact of experience on the riskiness of NTS holdings.

This table reports the results of cross-sectional regressions of the new-issue spreads on mutual funds' 2007 nontraditional securitization holdings on manager inexperience and controls:

$$\text{Spread}_i = \alpha_{\text{objective}(i)} + \beta_1 \cdot \text{Inexperienced}_i + \gamma' \mathbf{x}_i + \varepsilon_i.$$

In Panel A, the dependent variable is the par-weighted average new-issue spread on funds' 2007 NTS holdings. In Panel B, the new-issue spreads are deviations from the average spread for nontraditional securitizations with the same initial rating and issued in the same quarter. Fund objective fixed effects and the controls from Table 2 are included as indicated in the table. Robust t-statistics are shown in square brackets.

	(1)	(2)	(3)	(4)
<i>Panel A: Raw NTS spreads</i>				
<i>Inexperienced</i>	16.956 [2.73]	15.475 [2.44]	9.369 [1.82]	6.695 [1.38]
<i>Constant</i>	38.751 [13.55]	82.394 [3.15]	42.74 [13.25]	95.959 [3.71]
Controls	No	Yes	No	Yes
Fund objective fixed effects	No	No	Yes	Yes
Observations	350	350	350	350
R-squared	0.02	0.05	0.29	0.33
<i>Panel B: NTS spreads adjusted for rating and quarter-of-issue</i>				
<i>Inexperienced</i>	1.25 [0.49]	0.097 [0.04]	0.205 [0.09]	-1.382 [-0.57]
<i>Constant</i>	5.003 [2.82]	21.951 [2.11]	5.552 [3.06]	23.673 [2.04]
Controls	No	Yes	No	Yes
Fund objective fixed effects	No	No	Yes	Yes
Observations	350	350	350	350
R-squared	0.00	0.07	0.10	0.15

(a) Table 4: Riskiness of NTS holdings.

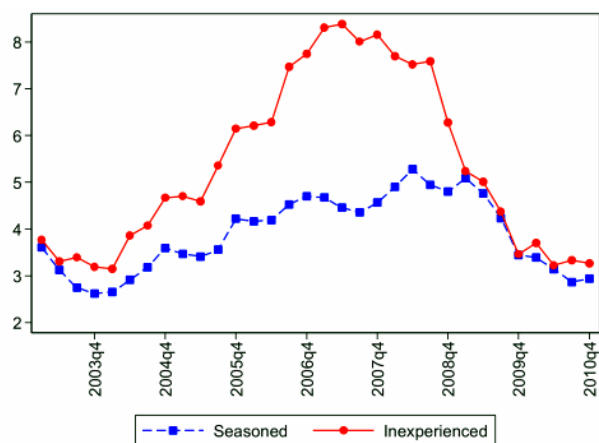


Fig. 3. Quarterly NTS holdings of seasoned and inexperienced bond fund managers. This figure shows the average nontraditional share of bond mutual funds managed by seasoned and inexperienced portfolio managers from 2003Q1 to 2010Q4. We split bond mutual funds into two groups based on the median value of the fund manager's experience measured as of 2004Q4. For team-managed funds, we take the average of individual managers' experience. Nontraditional share is total net holdings of

(b) Figure 3: Quarterly NTS holdings.

7.6 Table 5 and Figure 4: Evolution over time and local house price appreciation

Read:

- Panel A of Table 5 shows the inexperience effect growing from 2003 to 2007 and declining after the crisis.
- Panel B shows that high local house price appreciation predicts more NTS holdings, especially for inexperienced managers.
- Figure 4 plots fund assets by location and local house price appreciation, documenting substantial geographic variation in managers' local housing experiences.

Detailed interpretation: This table is the central availability-heuristic evidence. Managers in high-HPA locations personally experienced a strong housing boom. Inexperienced managers in such locations had the strongest NTS demand.

Economic message: Professional investors were influenced by local personal experiences, not just national data about housing.

Table 5

Investor experience and the evolution of NTS holdings over time.
Panel A reports separate cross-sectional regressions for $t = 2003$ to 2010 of bond mutual funds' holdings of nontraditional securitizations on their portfolio managers' experience and various fund characteristics:

$$NTS\ share_t = \alpha_{\text{experience}(t)} + \beta_1 \cdot \text{Inexperienced}_t + \beta_2 \cdot \text{High local HPA}_t + \gamma' \mathbf{X}_t + \varepsilon_t$$

The dependent variable in Panel A is the nontraditional share in year t . *Inexperienced* is a dummy indicator equal to one for managers with below-median experience as of 2004Q4 and does not vary over time for a given mutual fund. Panel B reports cross-sectional regressions of mutual funds' 2007 nontraditional share on portfolio manager experience and local house price appreciation:

$$NTS\ share_t = \alpha_{\text{experience}(t)} + \beta_1 \cdot \text{Inexperienced}_t + \beta_2 \cdot \text{High local HPA}_t + \beta_3 \cdot \text{High local HPA}_t \times \text{Inexperienced}_t + \gamma' \mathbf{X}_t + \varepsilon_t$$

Local house price appreciation is the annualized change in house prices from 2003 to 2006 for the MSA in which the investment manager is located, computed using the all-transactions index from the Federal Housing Finance Agency. *High local HPA* is an indicator for managers who experienced local HPA in the top tercile. The same controls as in Table 2 and objective fixed effects are included as indicated in the table below. t -statistics are shown in square brackets. Panel A uses robust t -statistics and Panel B uses t -statistics that allow for clustering at the MSA level.

Panel A: Experience and the evolution of NTS holdings from 2003 to 2010							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	2003	2004	2005	2006	2007	2008	2009
<i>Inexperienced</i>	0.325	0.703	1.035	1.794	3.376	2.169	-0.166
	[0.77]	[1.80]	[2.40]	[3.10]	[4.66]	[3.01]	[-0.25]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fund objective fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	767	797	818	785	743	710	631
R-squared	0.28	0.34	0.34	0.31	0.23	0.21	0.14
Panel B: Local house price appreciation and 2007 NTS holdings							
	(1)	(2)	(3)	(4)			
<i>Inexperienced</i>	3.422	2.001	3.228	1.865			
	[3.90]	[1.56]	[3.70]	[1.48]			
High local HPA	2.082	0.530	2.428	0.933			
	[2.64]	[0.71]	[2.58]	[1.20]			
High local HPA $\times$ <i>Inexperienced</i>		3.047		2.961			
		[2.00]		[1.88]			
Controls	Yes	Yes	Yes	Yes			
Objective fixed effects	No	No	Yes	Yes			
Observations	753	753	753	753			
R-squared	0.12	0.12	0.24	0.24			

(a) Table 5: Evolution and local HPA.

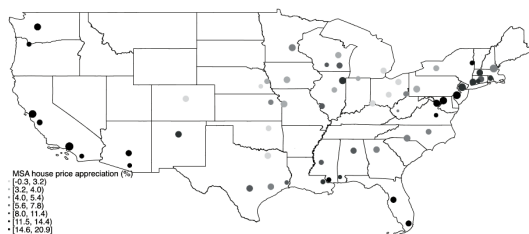


Fig. 4: Distribution of bond mutual fund assets and house price appreciation. This figure plots the distribution of bond mutual fund assets and house p

(b) Figure 4: Fund assets and house price appreciation.

7.7 Table 6: Prior manager return experiences and 2007 NTS holdings

Read:

- Inexperienced remains positive and significant across specifications.
- High average past returns do not have a stable independent effect.
- High minimum returns and distant minimum returns are positively related to NTS holdings.
- Managers whose worst experience was further in the past hold more NTS.

Detailed interpretation: The table shows that not all experience is equal. The salience and recency of bad outcomes matter. Managers who were not recently burned are more likely to buy NTS.

Economic message: Experience is path dependent. What managers personally lived through matters more than the mere number of years worked.

Table 6

The impact of prior manager return experiences on 2007 NTS holdings.

This table reports the results of cross-sectional regressions of mutual funds' 2007 nontraditional share on the past investment return outcomes experienced by the fund manager:

$$NTS\ share_t = \alpha_{\text{experience}(t)} + \beta_1 \cdot \text{Inexperienced}_t + \beta_2 \cdot \text{HighAvgReturn}_t + \beta_3 \cdot \text{HighMinReturn}_t + \beta_4 \cdot \text{DistantMinReturn}_t + \gamma' \mathbf{X}_t + \varepsilon_t$$

The dependent variable is the nontraditional share in 2007. *Inexperienced* is a dummy indicator equal to one for managers with below-median experience as of 2004Q4. We first compute each manager's average return from 1995 to 2005 (average across managers for team-managed funds). *HighAvgReturn* is an indicator for managers with average returns in the top two terciles. Similarly, we compute each manager's minimum return from 1995 to 2005 (average across managers for team-managed funds). *HighMinReturn* is an indicator for managers with minimum returns in the top two terciles. *DistantMinReturn* is a continuous measure that equals one if a manager's minimum return from 1995 to 2005 was in 1995, that equals zero if the minimum was in 2005, and so on. Formally, $\text{DistantMinReturn} = (2005 - \text{YearMin} - 1995) / (2005 - 1995)$. Fund objective fixed effects and the controls from Table 2 are included as indicated in the table. Robust t -statistics are shown in square brackets.

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Inexperienced</i>	3.576	3.093	3.745	3.431	3.087	3.698
	[4.58]	[3.92]	[4.46]	[4.73]	[4.12]	[4.56]
High average return	0.389	-0.808	-1.26	0.525	-0.29	-0.762
	[0.45]	[-0.79]	[-1.27]	[0.59]	[-0.27]	[-0.73]
High min return		2.696	2.815		1.908	2.191
		[2.94]	[3.05]		[1.84]	[2.12]
Distant min return			5.932			5.712
			[2.73]			[2.82]
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Objective fixed effects	No	No	No	Yes	Yes	Yes
Observations	755	755	755	755	755	755
R-squared	0.11	0.12	0.13	0.23	0.23	0.24

Table 6: Prior manager return experiences and 2007 NTS holdings.

7.8 Table 7 and Figure 5: 1998 credit-market experience

Read:

- Panel A shows that the effect of experience is not linear; the important break is around whether the manager lived through the 1998 credit disruption.
- Panel B shows that among managers active in 1998, good 1998 outcomes predict higher 2007 NTS holdings.
- Figure 5 visualizes the nonlinear cutoff pattern around the 1998 crisis.

Detailed interpretation: This is one of the most convincing mechanism tests. Managers who experienced the 1998 Russia/LTCM turmoil were less likely to neglect tail risk later. Within that group, those personally hurt in 1998 learned more.

Economic message: Financial memory is event-based, not clock-based. Crises teach risk more effectively than routine years of experience.

Table 7
Personal experiences during prior market disruptions: The impact of 1998 manager experiences on 2007 NTS holdings.
Panel A reports the results of cross-sectional regressions of mutual funds' 2007 nontraditional share on fund manager inexperience defined using different cutoff dates:

$$NTS_{share} = \alpha_{0(NTS_{share})} + \beta \cdot 1[Manager \text{ starts after } 1/1/YYYY] + \gamma'X + \epsilon_i$$

Panel B reports the results of cross-sectional regressions of mutual funds' 2007 nontraditional share on the 1998 investment outcome experienced by the manager:

$$NTS_{share} = \alpha_{0(NTS_{share})} + \beta_1 \cdot Inexperienced^{1998} + \beta_2 \cdot High \text{ outcome}^{1998} + \beta_3 \cdot Inexperienced^{1998} \times High \text{ outcome}^{1998} + \gamma'X + \epsilon_i$$

The sample of funds in Panel B consists of funds with at least one portfolio manager who managed a mutual fund during 1998. Fund manager's identity is fixed as of the end of 2004. $Inexperienced^{1998}$ is a dummy equal to one for managers above the median of the distribution of experience within the sample of managers who managed a fund during 1998. For each manager, we measure the minimum returns and fund flows she experienced across all funds she managed during 1998. For team-managed funds, we then take the average across all managers with 1998 experience. $High \text{ outcome}^{1998}$ is an indicator for funds whose 1998 outcome is in the top two terciles. All regressions include the same controls as the specifications reported in Table 2. Fund objective fixed effects are included as indicated in the table. Robust t-statistics are shown in square brackets.

Panel A: Manager experience does not accrue linearly over time								
	1995	1996	1997	1998	1999	2000	2001	2002
Manager starts after 1/1/YYYY	2.656	2.306	3.397	1.381	0.711	0.674	0.490	0.192
	[4.27]	[3.37]	[4.66]	[1.86]	[0.87]	[0.71]	[0.40]	[0.10]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Objective fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	757	757	757	757	757	757	757	757
R-squared	0.22	0.22	0.23	0.21	0.21	0.21	0.21	0.21
Panel B: Impact of personal 1998 investment outcomes								
	Outcome ¹⁹⁹⁸ = Returns ¹⁹⁹⁸				Outcome ¹⁹⁹⁸ = Flows ¹⁹⁹⁸			
Inexperienced ¹⁹⁹⁸	2.840	-0.449	2.882	-0.289	2.915	2.719	3.036	1.918
	[2.96]	[-0.37]	[3.08]	[-0.22]	[3.00]	[2.01]	[3.19]	[1.41]
High outcome ¹⁹⁹⁸	2.666	0.204	2.595	0.249	3.368	3.227	3.340	2.522
	[3.12]	[0.20]	[2.48]	[0.23]	[3.71]	[3.44]	[3.74]	[2.80]
High outcome ¹⁹⁹⁸ × Inexperienced ¹⁹⁹⁸		5.016	4.809	4.809		0.294		1.687
		[2.79]		[2.72]		[0.16]		[0.95]
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Objective fixed effects	No	No	Yes	Yes	No	No	Yes	Yes
Observations	493	493	493	493	487	487	487	487
R-squared	0.16	0.17	0.29	0.30	0.16	0.16	0.29	0.30

(a) Table 7: 1998 experience and 2007 NTS.

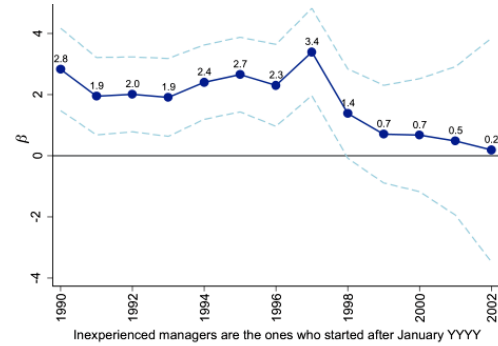


Fig. 5. Manager experience does not accrue linearly over time. This figure plots the coefficients β from estimating cross-sectional regressions of mutual funds' 2007 nontraditional share on fund manager inexperience, defined using different cutoff dates as in Panel A of Table 7:

$NTS_{share} = \alpha_{0(NTS_{share})} + \beta \cdot 1[Manager \text{ starts after } 1/1/YYYY] + \gamma'X + \epsilon_i$
(b) Figure 5: Experience does not accrue linearly.

7.9 Table 8: Institutional memory and manager training

Read:

- Panel A shows that family-level 1998 outcomes affect 2007 NTS holdings even after controlling for manager outcomes.
- The interaction terms suggest that inexperienced managers are influenced by institutional memory within the fund family.
- Panel B shows that CFA training is associated with lower NTS holdings, especially for inexperienced managers.

Detailed interpretation: Table 8 adds an organizational and educational layer. Personal experience is not the only source of memory. Fund families can transmit caution, and formal training can partially substitute for experience.

Economic message: Institutional memory and training can reduce the cyclicity of beliefs by preserving lessons from past crises.

Table 8

Substitutes for firsthand experience and 2007 NTS holdings: The roles of institutional memory and manager training.
Panel A reports cross-sectional regressions of mutual funds' 2007 nontraditional share on the 1998 investment outcome experienced by the fund manager and the investment outcome experienced by their fund family:

$$NTS\ share_t = \alpha_{Age(tenure)} + \beta_1 \cdot Inexperienced^{2008} + \beta_2 \cdot CFA + \beta_3 \cdot CFA \times Inexperienced + \gamma'X_t + \epsilon_t \\ + \beta_4 \cdot Inexperienced^{2008} \times High\ manager\ outcome^{2008} \\ + \beta_5 \cdot High\ family\ outcome^{2008} + \beta_6 \cdot Inexperienced^{2008} \times High\ family\ outcome^{2008} + \gamma'X_t + \epsilon_t$$

The sample in Panel A consists of funds with at least one manager who ran a mutual fund during 1998. Fund manager's identity is fixed as of the end of 2004. $Inexperienced^{2008}$ is a dummy equal to one for managers above the median of the distribution of experience among managers who managed a fund during 1998. As in Table 7, $High\ manager\ outcome^{2008}$ is an indicator for managers whose own 1998 outcome (returns or flows) is in the top two terciles. We compute the value-weighted outcome for all of the taxable bond funds in each family in 1998. $High\ family\ outcome^{2008}$ is an indicator for families whose 1998 outcome is in the top two terciles. Panel B reports cross-sectional regressions of funds' 2007 nontraditional share on manager experience and CFA status:

$$NTS\ share_t = \alpha_{Age(tenure)} + \beta_1 \cdot Inexperienced + \beta_2 \cdot CFA + \beta_3 \cdot CFA \times Inexperienced + \gamma'X_t + \epsilon_t$$

$Inexperienced$ manager is a dummy indicator equal to one for managers with below-median experience as of 2004Q4. CFA indicates whether the portfolio manager is a Chartered Financial Analyst (it is the fraction of CFA-chartered managers for a team-managed fund). Fund objective fixed effects and the controls from Table 2 are included as indicated in the table. Robust t-statistics are shown in square brackets.

Panel A: Institutional memory		Outcome ²⁰⁰⁸ = Returns ²⁰⁰⁸				Outcome ²⁰⁰⁸ = Flows ²⁰⁰⁸			
<i>Inexperienced²⁰⁰⁸</i>		3.185 [2.94]	-3.252 [-1.77]	3.303 [2.96]	-3.324 [-1.83]	2.656 [2.37]	0.966 [0.39]	2.894 [2.48]	-0.871 [-0.36]
<i>High manager outcome²⁰⁰⁸</i>		2.855 [2.66]	1.347 [1.05]	2.812 [2.26]	1.222 [1.00]	3.332 [3.21]	3.440 [3.30]	2.933 [2.87]	1.920 [1.98]
<i>High manager outcome²⁰⁰⁸ × Inexperienced²⁰⁰⁸</i>									
<i>High family outcome²⁰⁰⁸</i>		1.609 [1.51]	-1.840 [-1.42]	1.847 [1.78]	-1.730 [-1.52]	-0.259 [-0.23]	-1.664 [-1.52]	-0.186 [-0.17]	-1.952 [-1.51]
<i>High family outcome²⁰⁰⁸ × Inexperienced²⁰⁰⁸</i>									
<i>Controls</i>	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Objective × Quarter fixed effects</i>	No	No	Yes	Yes	Yes	No	Yes	Yes	Yes
<i>Observations</i>	383	383	383	383	412	412	412	412	412
<i>R-squared</i>	0.17	0.19	0.31	0.34	0.15	0.15	0.30	0.31	0.31
Panel B: Manager training									
<i>Inexperienced manager</i>		5.385 [3.63]		5.084 [3.65]		4.497 [3.08]		4.491 [3.25]	
<i>CFA</i>		-1.604 [-2.47]		-1.242 [-1.89]		-1.237 [-1.92]		-1.087 [-1.70]	
<i>CFA × Inexperienced</i>									
<i>Controls</i>	No	No	Yes	Yes	No	Yes	Yes	Yes	Yes
<i>Fund objective fixed effects</i>	No	No	Yes	Yes	No	Yes	Yes	Yes	Yes
<i>Observations</i>	757	757	757	757	757	757	757	757	757
<i>R-squared</i>	0.11	0.11	0.11	0.11	0.23	0.23	0.23	0.23	0.23

for each security. Using this data, we can decompose the total change in our NTS held by each fund into the cost flows in the crisis; or (ii) that all funds with high initial NTS holdings sold them during the crisis and experienced

Table 8: Substitutes for firsthand experience.

7.10 Table 9: Trading by inexperienced managers during the crisis

Read:

- During 2007Q3–2009Q2, inexperienced managers sold more NTS than seasoned managers.
- The result remains after controlling for outflows and lagged NTS share.
- Inexperienced managers' NTS also experienced larger passive declines from amortization and write-downs.

Detailed interpretation: Table 9 shows that inexperienced managers revised their views more dramatically once the boom broke. Their sales were not simply forced by investor redemptions; they sold more even after controls.

Economic message: Boom-time optimism turned into crisis-time retreat. The same managers who bought most aggressively sold aggressively when their beliefs were challenged.

7.11 Figure 6 and Table 10: Performance-flow incentives, returns, and alternative explanations

Read:

- Figure 6 shows that inexperienced managers did not have a stronger or more convex performance-flow relationship than seasoned managers.
- Table 10 shows that before the crisis inexperienced managers had similar returns and flows to seasoned managers.
- During the crisis, inexperienced managers had worse returns, but the effect is explained by greater NTS holdings.

Detailed interpretation: These results help rule out incentive-based alternatives. If inexperienced managers bought more NTS because they faced stronger risk-taking incentives, their performance-flow schedules should be more convex. They are not.

Table 9

Trading by inexperienced managers during the crisis: 2007Q3–2009Q2.

This table reports the results of regressions of mutual funds' quarterly trading behavior from 2007Q3 to 2009Q2 on manager inexperience, fund flows, and the lagged NTS share:

$$y_t = \alpha_{Age(tenure)} + \beta_1 \cdot Inexperienced + \beta_2 \cdot Flows_t + \beta_3 \cdot NTS\ Share_{t-1} + \gamma'X_t + \epsilon_t$$

In Panel A, the dependent variable is a measure of active NTS trading in each quarter:

$y_t = Par\ NTS\ sold - Par\ fixed-income\ holdings$. In Panel B, the dependent variable is a measure of passive NTS amortization in each quarter: $y_t = Decline\ in\ par\ NTS\ due\ to\ amortization - Par\ fixed-income\ holdings$. $Inexperienced$ is a dummy indicator equal to one for managers with below-median experience as of 2004Q4. Investment-objective-by-quarter fixed effects and the controls from Table 2 are included as indicated in the table. t-statistics that are robust to clustering at both the fund level and by quarter are shown in square brackets.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: $y_t = Par\ NTS\ sold - Par\ fixed-income\ holdings$						
<i>Inexperienced</i>	0.479 [2.27]	0.410 [2.22]	0.247 [1.80]	0.470 [2.06]	0.409 [2.03]	0.225 [1.48]
<i>Fund outflows</i>			0.158 [3.27]			0.161 [3.35]
<i>Lagged NTS share</i>			0.060 [1.24]			0.065 [1.29]
<i>Controls</i>	No	Yes	Yes	No	Yes	Yes
<i>Objective × Quarter fixed effects</i>	No	No	No	Yes	Yes	Yes
<i>Observations</i>	5,368	5,368	5,368	5,368	5,368	5,368
<i>R-squared</i>	0.00	0.01	0.04	0.05	0.05	0.08
Panel B: $y_t = Decline\ in\ par\ NTS\ due\ to\ amortization - Par\ fixed-income\ holdings$						
<i>Inexperienced manager</i>	0.275 [2.55]	0.254 [2.32]	0.251 [2.07]	0.243 [2.29]	0.084 [2.14]	0.084 [2.14]
<i>Fund outflows</i>			0.027 [2.76]			0.023 [2.26]
<i>Lagged NTS share</i>			0.073 [3.75]			0.073 [3.42]
<i>Controls</i>	No	Yes	Yes	No	Yes	Yes
<i>Objective × Quarter fixed effects</i>	No	No	No	Yes	Yes	Yes
<i>Observations</i>	5,368	5,368	5,368	5,368	5,368	5,368
<i>R-squared</i>	0.01	0.02	0.20	0.07	0.08	0.23

Table 9: Trading by inexperienced managers during the crisis.

Economic message: The evidence points to beliefs shaped by experience, not simply agency incentives or managerial skill differences.

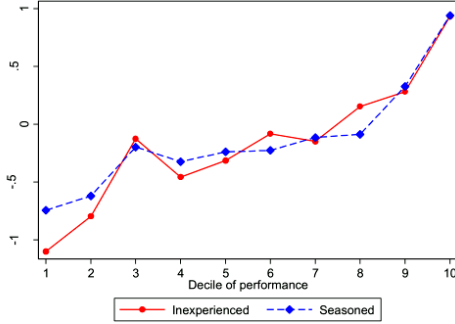


Fig. 6. Performance-flow relationship for seasoned and inexperienced bond fund managers. This figure reports the strength of the performance-

(a) Figure 6: Performance-flow relationship.

Table 10
Flows and returns: The role of investor experience and NTS holdings

Panel A reports panel regressions of returns on portfolio managers' experience, their holdings of NTS, and fund characteristics:

$$R_{it} = \alpha_{\text{experience}(i)} + \beta_1 \text{Inexperienced}_i + \beta_2 \text{NTS}_{i,t-1} + \beta_3 R_{i,t-1} + \beta_4 \text{FLOW}_{i,t-1} + \gamma' X_{i,t-1} + \epsilon_{it}$$

Panel B reports panel regressions of flows on their portfolio managers' experience, their holdings of NTS, and various fund characteristics:

$$\text{FLOW}_{i,t} = \alpha_{\text{experience}(i)} + \beta_1 \text{Inexperienced}_i + \beta_2 \text{NTS}_{i,t-1} + \beta_3 R_{i,t-1} + \beta_4 \text{FLOW}_{i,t-1} + \gamma' X_{i,t-1} + \epsilon_{it}$$

Inexperienced is a dummy indicator equal to one for managers with below-median experience as of 2004Q4. Investment-objective-by-quarter fixed effects and the controls from Table 2 are included in all specifications. t-statistics that are robust to clustering at both the fund level and by quarter are shown in square brackets.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Returns												
	Pre-crisis (2003Q1–2007Q2)				Crisis (2007Q3–2009Q2)							
Inexperienced _{it}	-0.042 [-1.54]	-0.041 [-1.53]	-0.038 [-1.64]	-0.035 [-1.63]	-0.310 [-2.82]	-0.156 [-1.54]	-0.231 [-2.92]	-0.141 [-1.50]				
NTS _{it-1}	-0.001 [-0.88]	-0.001 [-0.77]	-0.001 [-0.86]	-0.001 [-0.74]	-0.073 [-2.84]	-0.072 [-2.80]	-0.049 [-1.52]	-0.048 [-1.42]				
R _{it-1}			0.129 [1.50]	0.129 [1.51]					0.361 [2.04]	0.311 [1.88]	0.311 [1.88]	
FLOW _{it-1}			-0.009 [-0.19]	-0.009 [-0.17]					0.032 [0.57]	0.019 [0.37]	0.018 [0.36]	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Objective-Quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	11,441	11,441	11,441	11,441	5,310	5,310	5,310	5,310	5,310	5,310	5,310	
R-squared	0.88	0.88	0.88	0.88	0.88	0.76	0.78	0.78	0.79	0.79	0.79	
Panel B: Flows												
	Pre-crisis (2003Q1–2007Q2)				Crisis (2007Q3–2009Q2)							
Inexperienced _{it}	-0.101 [-1.17]	-0.113 [-1.34]	-0.084 [-1.27]	-0.094 [-1.45]	-0.295 [-2.21]	-0.204 [-1.56]	-0.177 [-1.96]	-0.140 [-1.47]				
NTS _{it-1}	0.010 [1.55]	0.011 [1.65]	0.008 [1.27]	0.009 [1.35]	-0.044 [-3.80]	-0.043 [-3.83]	-0.021 [-2.76]	-0.020 [-2.59]				
R _{it-1}			0.259 [5.30]	0.261 [5.22]					0.147 [5.19]	0.126 [4.51]	0.126 [4.52]	
FLOW _{it-1}			0.237 [8.50]	0.237 [8.44]					0.171 [7.27]	0.166 [7.24]	0.165 [7.21]	
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Objective-Quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(b) Table 10: Flows and returns.

8 Detailed result map

- **Boom context:** NTS issuance exploded, and NTS spreads compressed during the boom.
- **Baseline result:** inexperienced managers held far more NTS in 2007.
- **Risk result:** inexperienced managers bought riskier NTS with higher spreads/lower ratings.
- **Time-series result:** the experience gap developed during the tranquil boom and narrowed after the crisis.
- **Local experience result:** high local house price appreciation increased NTS demand, especially for inexperienced managers.
- **Personal outcome result:** managers who avoided severe or recent bad returns bought more NTS.
- **1998 experience result:** managers active during the Russia/LTCM disruption bought less NTS in 2007, especially if personally hurt.
- **Substitute result:** fund family memory and CFA training partially substitute for firsthand experience.
- **Crisis behavior result:** inexperienced managers sold more and suffered larger NTS write-downs during the crisis.

9 Short summary

- The paper studies whether personal investment experience shaped demand for risky securitizations during the mortgage credit boom.
- It uses mutual fund holdings and portfolio manager biographies to measure both NTS exposure and manager experience.
- Inexperienced managers held far more NTS than seasoned managers in 2007.
- They also held riskier NTS and increased exposure more during the boom.
- Local house price appreciation, favorable personal outcomes, and the absence of 1998 crisis experience all predict greater NTS demand.
- Institutional memory and CFA training reduce NTS holdings among inexperienced managers.

- During the crisis, inexperienced managers sold more NTS and held securities that were written down more severely.
- The evidence supports a path-dependent view of credit booms: tranquility breeds optimistic beliefs, especially among investors without firsthand bad experiences.

10 Conclusion

10.1 Core conclusion

Chernenko, Hanson, and Sunderam show that investor experience was a powerful determinant of who bought risky mortgage-linked securitizations during the credit boom. Inexperienced fund managers, especially those lacking salient negative experiences, bought substantially more nontraditional securitizations than seasoned managers.

10.2 Incremental contribution revisited

The paper’s incremental contribution is to connect the macro narrative of credit booms to micro evidence on professional investors. It shows that boom-time optimism was not uniform; it was strongest among managers whose personal histories made downside risk less salient. It also shows that local experiences, past returns, 1998 crisis exposure, institutional memory, and training all shaped risk taking.

10.3 Policy implication

The paper suggests a nonstandard lesson for financial stability. Having investors bear moderate losses may preserve financial memory. If policymakers always suppress downside realizations, they may unintentionally encourage future investors to neglect tail risk.

10.4 Limitations

The setting is mutual funds rather than banks or hedge funds, and experience is measured from available manager histories. Holdings data measure portfolios but not all beliefs directly. Nonetheless, the consistency across holdings, riskiness, local experience, past outcomes, 1998 exposure, training, and crisis trading makes the interpretation persuasive.

10.5 Takeaway

The central takeaway is that financial memory fades. As generations of investors without firsthand crisis experience take more responsibility, credit booms can become more fragile because downside risk becomes less available in investors’ beliefs.